CYT and SKT systems for invariant metrics; SKT+CYT implies Bismut flat (Corollary 5.1)

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/sigma-2023-028-cyt-and-skt-metrics-on-compact-semi-simple-l/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span is the author Corollary environment coroBismutflat with its own label and its own single proof environment (source lines 448-540). The author's phrase 'the last part of the theorem' at source line 476 (for the corollary's part (iii)) is copied verbatim.

Editorial note. The Lambda definition (source lines 377-379) is delivered as a context span before the statement; the two sibling statements (Theorem 4.2, Corollary 3.4) are delivered as context spans after the statement so \ref{th4.1} and \ref{th4.1bis} resolve and the card keeps 5.1 (both print locally renumbered).

Editorial note. No bibliography fragment is needed: the proof span carries no \cite commands.

Editorial note. The pinned source's stray Unicode non-breaking space U+00A0 (source line 451, inside the delivered proof span) is mapped to an ordinary space in the card preamble via newunicodechar; the delivered span text itself remains a verbatim line copy of the pinned source and no source byte is modified.

Editorial note. Printed number anchored to the published PDF (SIGMA 19 (2023) 028, read from the stream-extracted text of sigma23-028.pdf downloaded from sigma-journal.com). The pinned tex has five \section commands and no \appendix and its section structure matches the published PDF one-to-one, so the unit 010 pinned-versus-published realignment does not trigger.

Editorial note. Equation and theorem numbers inside the card are locally renumbered; every cross-reference is a symbolic \eqref/\ref and resolves inside the card.

Editorial scope. Single environment card cut from the SIGMA article doi:10.3842/SIGMA.2023.028 (sigma-2023-028-cyt-and-skt-metrics-on-compact-semi-simple-l); statement and proof are the author's own text at source lines 430-445 and 448-540.

Allowed standard prerequisites (editorial).

- CYT characterization (Theorem 4.2). Delivered in context (source lines 420-424): g is CYT iff sum (d alpha - d(alpha circle Lambda)/lambda_alpha) = 0.. Load-bearing: part (i) translates the CYT condition into the system (CYTsystem) via the simple-root expansion of alpha.
- SKT criterion (Corollary 3.4). Delivered in context (source lines 214-220): the SKT criterion in root coordinates.. Load-bearing: part (ii) translates the SKT condition into the linear system (SKTsystem).
- Simple-root data. Standard facts used by the author in-span: d alpha_j = sum B(H_alpha_j, H_alpha) e wedge Ie (source lines 466-469), B(H, H_alpha) = alpha(H) (source lines 513-519), and the r x p simple-root coefficient matrix (x_alpha^i) has rank r (source lines 534-538).. Load-bearing: the expansion of part (ii) and the invertibility argument of part (iii).

Let g be a left-invariant I-Hermitian metric on G which is also invariant by the right T-action. Then we can write $g(\cdot,\cdot) = -B(\Lambda(\cdot),\cdot)$, where Λ is a positive-definite Hermitian matrix on $\mathfrak{t}$ and diagonal on $\sum_\alpha \mathfrak{g}_\alpha$. In particular, $g(E_\alpha, E_{-\alpha}) = \lambda_\alpha$ and $g(E_\alpha, E_\beta) = 0$, for every $\alpha \neq -\beta$.

Corollary 5.1. *Let G be a compact semi-simple Lie group with Samelson's complex structure I determined by the set P of positive roots associated to a maximal torus T of G. Let g be a left-invariant I-Hermitian metric on G, which is also invariant under the right action of the torus T in the Tits fibration $G \to G/T$. Let $\alpha_1, \dots, \alpha_r$ be the choice of simple roots and p be the cardinality of P. Then*

(i) g is CYT if and only if for every $\alpha \in P$ there exist p positive real numbers $\lambda_\alpha > 0$, $\alpha \in P$, satisfying the r equations

$$\sum_{\alpha\in P}\left(x^i_\alpha - \sum_{j=1}^{r}\frac{x^j_\alpha}{\lambda_\alpha}\Lambda^i_j\right) = 0, \qquad i = 1,\dots,r, \tag{5.1}$$

where x^j_α denotes the coefficient of α with respect to the simple root α_j and Λ^j_i are the entries of Λ in the basis $(\alpha_1,\dots,\alpha_r)$.

(ii) g is SKT if and only if there exist r real numbers $\tilde{N}_j$ and p positive real numbers $\tilde{\lambda}_\alpha > 0$, $\alpha \in P$, satisfying the p equations

$$\tilde{\lambda}_\alpha - 1 - \sum_{j=1}^{r}\tilde{N}_j B(H_{\alpha_j}, H_\alpha) = 0, \qquad \alpha \in P. \tag{5.2}$$

(iii) If g is both SKT and CYT, then g is Bismut flat.

Theorem 5.2. *Let G be a compact semi-simple Lie group with a Samelson's complex structure determined by the set P of positive roots associated to a maximal torus T of G. Let g be a left-invariant I-Hermitian metric on G, which is also invariant under the right action of the torus T in the Tits fibration $G \to G/T$, where T is a maximal torus in G. We can write the fundamental form F of the metric g as $F = F|_{\mathfrak{t}} + \sum_{i=1}^{m}\sum_{\alpha\in P_i}\lambda^{(i)}_\alpha e_\alpha \wedge Ie^{(i)}_\alpha$, where $e^{(i)}_\alpha$, $Ie^{(i)}_\alpha$ are dual to X_α, Y_α and $\lambda^{(i)}_\alpha = g(E_\alpha, E_{-\alpha})$. Then g is CYT iff $\sum_{\alpha\in P}\big(\mathrm{d}\alpha - \frac{1}{\lambda_\alpha}\mathrm{d}(\alpha\circ\Lambda)\big) = 0$, where Λ is the positive operator such that $g|_{\mathfrak{t}}(\cdot,\cdot) = -B(\Lambda(\cdot),\cdot)$.*

Corollary 5.3. *Let G be a compact semi-simple Lie group with Samelson's complex structure I as above. Let g be a left-invariant I-Hermitian metric on G, which is also invariant under the right action of the torus T in the Tits fibration $G \to G/T$, where T is a maximal torus in G. We can write the fundamental form F of the metric g as $F = F|_{\mathfrak{t}} + \sum_{i=1}^{m} \sum_{\alpha \in P_i} \lambda_\alpha^{(i)} e_\alpha^{(i)} \wedge I e_\alpha^{(i)}$, where $e_\alpha^{(i)}$, $Ie_\alpha^{(i)}$ are dual to $X_\alpha^{(i)}, Y_\alpha^{(i)}$ and $\lambda_\alpha^{(i)} = g(E_\alpha^{(i)}, E_{-\alpha}^{(i)})$. Then g is SKT iff $g|_{\mathfrak{t}} = -\sum_{i=1}^{m} \nu_i B|_{\mathfrak{t}_i}$ and $\sum_{i=1}^{m} \sum_{\alpha \in P_i} \big(\lambda_\alpha^{(i)} - \nu_i\big) e_\alpha^{(i)} \wedge I e_\alpha^{(i)}$ is a linear combination of the basis $\omega_j = \mathrm{d}\alpha_j$ of $H^2(G/T, \mathbb{R})$, where α_j are the simple roots.*

Equivalently, g is SKT iff $\mathrm{d}\big(\sum_{i=1}^{m} \sum_{\alpha \in P_i} \big(\lambda_\alpha^{(i)} - \nu_i\big) e_\alpha^{(i)} \wedge I e_\alpha^{(i)}\big) = 0$ and $\Lambda|_{\mathfrak{t}} = \sum_{i=1}^{m} \nu_i \mathrm{Id}|_{\mathfrak{t}_i}$, for some constants ν_i, $i = 1, 2, \ldots, m$.

Proof. From Theorem 5.2, it follows that g is CYT if and only if there exist p positive real number λ_α, $\alpha \in P$, such that

$$\sum_{\alpha \in P} \left(\mathrm{d}\alpha - \frac{1}{\lambda_\alpha} \mathrm{d}(\alpha \circ \Lambda) \right) = 0.$$

Therefore, since $\alpha = \sum_{i=1}^{r} x_\alpha^i \alpha_i$, with x_α^i positive integers and Λ_i^j the entries of Λ, we have that the CYT condition is equivalent to

$$\sum_{\alpha \in P} \left(x_\alpha^i - \sum_{j=1}^{r} \frac{x_\alpha^j}{\lambda_\alpha} \Lambda_j^i \right) \mathrm{d}\alpha_i = 0.$$

Since the forms $\mathrm{d}\alpha_i$, $i = 1, \ldots, r$, are linearly independent, we get the non-linear system (5.1) of r equations in the p variables λ_α, $\alpha \in P$, and (i) follows.

Let denote by P_i the set of positive roots of the simple factor G_i with respect to $\mathfrak{t}_i$. To prove the second part of the corollary we use that by Corollary 5.3 the SKT condition is equivalent to the existence of r real numbers N_i, $i = 1, \ldots, r$ and p positive numbers λ_α, $\alpha \in P$, such that

$$\sum_{i=1}^{m} \sum_{\alpha \in P_i} \big(\lambda_\alpha^{(i)} - \nu_i\big) e_\alpha^{(i)} \wedge I e_\alpha^{(i)} = \sum_{i=1}^{r} N_i \, \mathrm{d}\alpha_i$$

and $\Lambda = \sum_{i=1}^{m} \nu_i \mathrm{Id}|_{\mathfrak{t}_i}$. Since

$$\mathrm{d}\alpha_j = \sum_{i=1}^{m} \sum_{\alpha \in P_i} B\big(H_{\alpha_j}, H_\alpha^{(i)}\big) e_\alpha^{(i)} \wedge I e_\alpha^{(i)}, \qquad j = 1, \ldots, r,$$

we obtain

$$\sum_{i=1}^{m} \sum_{\alpha \in P_i} \big(\lambda_\alpha^{(i)} - \nu_i\big) e_\alpha^{(i)} \wedge I e_\alpha^{(i)} = \sum_{i=1}^{m} \sum_{j=1}^{r} N_j \sum_{\alpha \in P_i} B\big(H_{\alpha_j}, H_\alpha^{(i)}\big) e_\alpha^{(i)} \wedge I e_\alpha^{(i)}.$$

Therefore, using that the previous equality holds on every simple factor $\mathfrak{g}_i$ and rescaling $\lambda_\alpha^{(i)}$ and N_i by ν_i we get the linear system (5.2) of p equations in the $p + r$ variables $\tilde{N}_i$, $i = 1, \ldots, r$, and $\tilde{\lambda}_\alpha$, $\alpha \in P$.

Now to prove the last part of the theorem suppose that g is both CYT and SKT. Then using that $\Lambda = \sum_{i=1}^{m} \nu_i \mathrm{Id}|_{\mathfrak{t}_i}$ and rescaling λ_α in (5.1) in the same

way as for the SKT case we have that the equations for the CYT condition become

$$\sum_{\alpha\in P} x_\alpha^i\left(1-\sum_{j=1}^r\frac{1}{\tilde\lambda_\alpha}\right)=0,\qquad i=1,\dots,r.$$

Let

$$P=\{\alpha_1,\dots,\alpha_r,\alpha_{r+1},\dots,\alpha_p\},$$

where $\alpha_1,\dots,\alpha_r$ are the simple roots. Imposing the SKT and CYT condition we get the following non-linear system of $p+r$ equations in the variables $\tilde\lambda_j$, $j=1,\dots,p$, $\tilde N_i$, $i=1,\dots,r$,

$$\begin{cases}\tilde\lambda_j-1-\sum_{i=1}^r\tilde N_i B(H_{\alpha_i},H_{\alpha_j})=0,\qquad j=1,\dots,p,\\ \sum_{j=1}^p x_{\alpha_j}^i\left(1-\frac{1}{\tilde\lambda_j}\right)=0,\qquad i=1,\dots,r.\end{cases}\tag{5.3}$$

Note that the solution $\tilde N_i=0$, $i=1,\dots,r$, $\tilde\lambda_j=1$, $j=1,\dots,p$, corresponds to a Bismut flat metric. Since $\tilde\lambda_j\neq 0$, for every $j=1,\dots,p$, if we multiply

$$\sum_{j=1}^p x_{\alpha_j}^i\left(1-\frac{1}{\tilde\lambda_j}\right)=0,\qquad i=1,\dots,r,$$

by the product of all the $\tilde\lambda_j$,we get

$$\sum_{j=1}^p x_{\alpha_j}^i\tilde\lambda_1\cdots\hat{\tilde\lambda}_j\cdots\tilde\lambda_p\big(\tilde\lambda_j-1\big)=0,\qquad i=1,\dots,r,$$

where $\hat{\tilde\lambda}_j$ denotes that the element $\tilde\lambda_j$ is removed in the product. We know that

$$\tilde\lambda_j-1=\sum_{k=1}^r\tilde N_k B(H_{\alpha_k},H_{\alpha_j})$$

for every $j=1,\dots,p$. Therefore

$$\sum_{j=1}^p x_{\alpha_j}^i\tilde\lambda_1\cdots\hat{\tilde\lambda}_j\cdots\tilde\lambda_p\left(\sum_{k=1}^r\tilde N_k B(H_{\alpha_k},H_{\alpha_j})\right)=0,\qquad i=1,\dots,r,$$

which can be viewed as a homogeneous system of r equations in the r variables $\tilde N_i$, $i=1,\dots,r$. The $r\times r$ associated matrix $M=M\big(\tilde\lambda_1,\dots,\tilde\lambda_p\big)$ has (i,k)-th entry given by

$$\sum_{j=1}^p x_{\alpha_j}^i\tilde\lambda_1\cdots\hat{\tilde\lambda}_j\cdots\tilde\lambda_p B(H_{\alpha_k},H_{\alpha_j}).$$

We can rewrite $B(H_{\alpha_k},H_{\alpha_j})$, $k=1,\dots,r$, $j=1,\dots,p$, in terms of $B(H_{\alpha_k},H_{\alpha_j})$ with $k,j\in\{1,\dots,r\}$, using that $B(H,H_\alpha)=\alpha(H)$ for every $H\in\mathfrak{t}$. Indeed, if $\alpha=\sum_{l=1}^r x_\alpha^l\alpha_l$ we have,

$$B(H_{\alpha_i},H_\alpha)=\alpha(H_{\alpha_i})=\sum_{l=1}^r x_\alpha^l\alpha_l(H_{\alpha_i})=\sum_{l=1}^r x_\alpha^l B(H_{\alpha_i},H_{\alpha_l}),$$

for every $i = 1, \dots, r$. As consequence the (i,k)-th entry of the matrix $M = M\big(\tilde{\lambda}_1, \dots, \tilde{\lambda}_p\big)$ can be written as

$$\begin{aligned}\sum_{j=1}^{p} x^i_{\alpha_j} \tilde{\lambda}_1 \cdots \hat{\tilde{\lambda}_j} \cdots \tilde{\lambda}_p B(H_{\alpha_k}, H_{\alpha_j}) &= \sum_{j=1}^{p} x^i_{\alpha_j} \tilde{\lambda}_1 \cdots \hat{\tilde{\lambda}_j} \cdots \tilde{\lambda}_p B(H_{\alpha_j}, H_{\alpha_k}) \\ &= \sum_{j=1}^{p} x^i_{\alpha_j} \tilde{\lambda}_1 \cdots \hat{\tilde{\lambda}_j} \cdots \tilde{\lambda}_p \sum_{l=1}^{r} x^l_{\alpha_j} B(H_{\alpha_l}, H_{\alpha_k}) \\ &= B\bigg(\sum_{l=1}^{r} \sum_{j=1}^{p} x^i_{\alpha_j} \tilde{\lambda}_1 \cdots \hat{\tilde{\lambda}_j} \cdots \tilde{\lambda}_p x^l_{\alpha_j} H_{\alpha_l}, H_{\alpha_k} \bigg),\end{aligned}$$

where $i, k = 1, \dots, r$. Since

$$\sum_{j=1}^{p} x^i_{\alpha_j} \tilde{\lambda}_1 \cdots \hat{\tilde{\lambda}_j} \cdots \tilde{\lambda}_p x^l_{\alpha_j} = \tilde{\lambda}_1 \cdots \tilde{\lambda}_p \sum_{j=1}^{p} x^i_{\alpha_j} \frac{1}{\tilde{\lambda}_j} x^l_{\alpha_j},$$

it will be sufficient to show, for every $\tilde{\lambda}_j$, the invertibility of the matrix L, whose (i,l)-entry is given

$$\sum_{j=1}^{p} x^i_{\alpha_j} \tilde{\lambda}_1 \cdots \hat{\tilde{\lambda}_j} \cdots \tilde{\lambda}_p x^l_{\alpha_j} = \tilde{\lambda}_1 \cdots \tilde{\lambda}_p \sum_{j=1}^{p} x^i_{\alpha_j} \frac{1}{\tilde{\lambda}_j} x^l_{\alpha_j}.$$

Since the $r \times p$ matrix (x^i_α) has rank r, the r vectors

$$v_1 = \big(x^1_{\alpha_1}, \dots, x^1_{\alpha_p}\big), \quad \dots, \quad v_r = \big(x^r_{\alpha_1}, \dots, x^r_{\alpha_p}\big),$$

are linearly independent. Using that for every $j = 1, \dots, p$, $\tilde{\lambda}_j > 0$, we get that $\sum_{j=1}^{p} \frac{1}{\tilde{\lambda}_j} x^i_{\alpha_j} x^l_{\alpha_j}$ can be viewed as a positive definite scalar product on $\mathbb{R}^p$ with weights $\frac{1}{\tilde{\lambda}_j}$ of the two vectors v_i and v_l. Therefore, the matrix L is invertible and $\tilde{N}_i = 0$, for every $i = 1, \dots, r$. By (5.3), it follows that $\tilde{\lambda}_j = 1$, for every $j = 1, \dots, p$. □